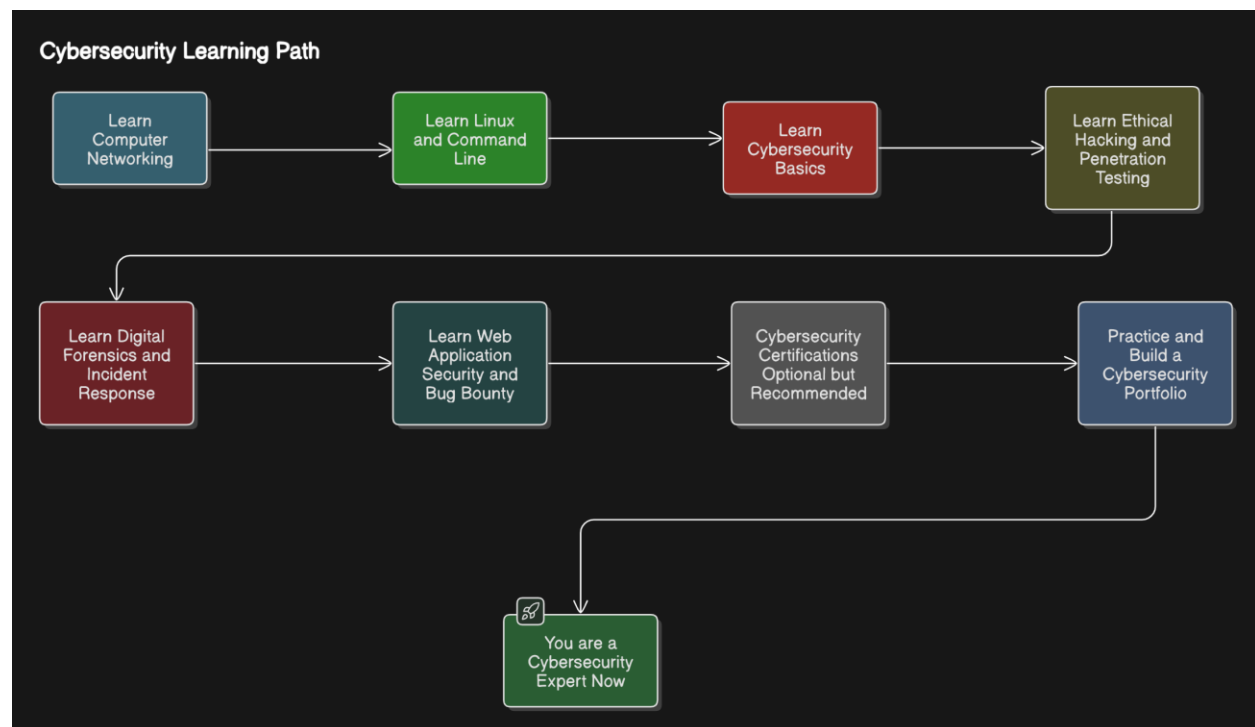


Detailed Roadmap & Free Resources to Learn Cybersecurity

Detailed Roadmap & Free Resources to Learn Cybersecurity

This roadmap is designed for complete beginners to acquire cybersecurity skills, even without any prior coding or computer science knowledge. It covers all technical skills with free learning resources provided for each.

Expected Duration to learn Cybersecurity: 6 to 9 months



How to Become a Cybersecurity Expert

Step 1 - Learn Computer Networking

Topics you need to cover in Networking for Cybersecurity:

- OSI and TCP/IP Models
- IP Addressing & Subnetting
- DNS, DHCP, NAT
- VPNs, Firewalls, and Proxies
- HTTP, HTTPS, FTP, SMTP Protocols
- Network Security Concepts (IDS, IPS, VLAN, DMZ)
- Packet Analysis using Wireshark

YouTube Videos to Learn Networking:

- [Computer Networking Full Course](#)
- [Networking Fundamentals](#)

Free Courses to Learn Networking:

- [Cisco Networking Academy](#)
- [Google IT Support](#)
- [Learn Networking Basics](#)

Step 2 - Learn Linux & Command Line

Topics to Cover in Linux:

- Linux File System & Permissions
- Basic Shell Commands
- Process Management
- Users, Groups, and Permissions
- Networking in Linux
- Bash Scripting

YouTube Videos to Learn Linux:

- [Linux for Beginners](#)
- [Linux Command Line](#)

Free Courses to Learn Linux:

- [Introduction to Linux - edX](#)
- [Linux Fundamentals - Cybrary](#)

Step 3 - Learn Cybersecurity Basics

Topics to Cover in Cybersecurity:

- Types of Cyber Attacks (Phishing, DoS, SQL Injection, XSS)
- Basics of Cryptography
- Authentication & Authorization
- Security Policies & Frameworks
- Incident Response & Threat Intelligence
- Web Security & OWASP Top 10

YouTube Videos to Learn Cybersecurity:

- [Introduction to Cybersecurity](#)
- [OWASP Top 10 Explained](#)

Free Courses to Learn Cybersecurity:

- [Google Cybersecurity Certification](#)
- [Cybrary Free Cybersecurity Training](#)
- [Cisco Introduction to Cybersecurity](#)

Step 4 - Learn Ethical Hacking & Penetration Testing

Topics to Cover in Ethical Hacking:

- Footprinting & Reconnaissance
- Scanning & Enumeration
- Exploiting Vulnerabilities

- Privilege Escalation
- Maintaining Access
- Covering Tracks

YouTube Videos to Learn Ethical Hacking:

- [Ethical Hacking Full Course](#)
- [Nmap for Beginners](#)

Free Courses to Learn Ethical Hacking:

- [TryHackMe \(Beginner to Advanced\)](#)
- [Hacker101](#)
- [OWASP Web Security Academy](#)

Step 5 - Learn Digital Forensics & Incident Response (DFIR)

Topics to Cover in DFIR:

- Windows & Linux Forensics
- Memory Analysis
- Log Analysis
- Malware Analysis Basics
- Reverse Engineering
- Chain of Custody & Legal Considerations

YouTube Videos to Learn Digital Forensics:

- [Introduction to Digital Forensics](#)
- [Memory Forensics using Volatility](#)

Free Courses to Learn Digital Forensics:

- [DFIR Training - Cybrary](#)
- [SANS Digital Forensics](#)

Step 6 - Learn Web Application Security & Bug Bounty

Topics to Cover:

- OWASP Top 10 Vulnerabilities
- Web Penetration Testing
- SQL Injection, XSS, CSRF
- API Security
- Burp Suite & ZAP Proxy

YouTube Videos to Learn Web Security:

- [Bug Bounty Hunting Full Course](#)
- [Web App Penetration Testing](#)

Free Platforms for Bug Bounty Practice:

- [HackerOne](#)
- [Bugcrowd](#)
- [PentesterLab](#)

Step 7 - Cybersecurity Certifications (Optional but Recommended)

Certifications to Boost Your Cybersecurity Career:

- **CompTIA Security+** (Best for Beginners) - [Learn More](#)
- **Certified Ethical Hacker (CEH)** - [Learn More](#)
- **Certified Information Systems Security Professional (CISSP)** - [Learn More](#)
- **Offensive Security Certified Professional (OSCP)** - [Learn More](#)

Step 8 - Practice & Build a Cybersecurity Portfolio

Hands-on Practice Platforms:

- [TryHackMe](#)

- [Hack The Box](#)
- [PentesterLab](#)
- [OverTheWire](#)

Contribute to Open-Source Security Projects:

- [GitHub Security Projects](#)
- [CVE Databases & Reporting](#)

Start a Cybersecurity Blog or YouTube Channel

- Document your learning journey, write articles, and share knowledge!

Final Notes:

By following this roadmap and utilizing the free resources, you can develop a strong foundation in cybersecurity and even start working in the field within **6 to 9 months**. The key is **consistent practice and hands-on experience**.

Happy Learning! Stay Safe Online!

More Comprehensive Cybersecurity Learning & Resources

GitHub Resources

1. **PayloadsAllTheThings** – A repository with payloads for pentesting & security testing

 <https://github.com/swisskyrepo/PayloadsAllTheThings>

2. **SecLists** – Security-related wordlists for pentesting

 <https://github.com/danielmiessler/SecLists>

3. **Resources for Beginner Bug Bounty Hunters** – Beginner-friendly bug bounty resources

 <https://github.com/nahamsec/Resources-for-Beginner-Bug-Bounty-Hunters>

4. **Awesome Bug Bounty** – A curated list of bug bounty resources

 <https://github.com/djadmin/awesome-bug-bounty>

5. **Public Bug Bounty Programs** – List of public bug bounty programs

 <https://github.com/projectdiscovery/public-bugbounty-programs>

Cybersecurity Project Ideas

1. **Beginner-Friendly AI-based Cybersecurity Projects**

 <https://analyticsindiamag.com/ai-trends/5-cybersecurity-projects-for-beginners/>

2. **ResearchGate Cybersecurity Project Discussions**

 https://www.researchgate.net/post/Does_anyone_has_any_Cyber_Security_related_project_idea_topic

3. **Top Cybersecurity Project Ideas (GeeksforGeeks)**

 <https://www.geeksforgeeks.org/top-6-cybersecurity-projects-ideas-for-beginners/>

4. **Jigsaw Academy's Cybersecurity Training & Projects**

 <https://jigsawacademy.com/>

5. **Information Security Projects (NevonProjects)**

 <https://nevonprojects.com/information-security-projects/>

6. **Cybersecurity Project Topics (UpGrad)**

 <https://www.upgrad.com/blog/cyber-security-project-ideas-topics/>

Cybersecurity Certifications

Entry-Level

1. **CompTIA Security+** – Covers basic cybersecurity knowledge

 <https://www.comptia.org/certifications/security>

2. **Certified Ethical Hacker (CEH) - EC-Council**

 <https://www.eccouncil.org/train-certify/certified-ethical-hacker-ceh/>

Intermediate & Advanced

1. **GIAC Penetration Tester (GPEN) - Advanced Pentesting**

 <https://www.giac.org/certifications/penetration-tester-gpen/>

2. **Offensive Security Certified Professional (OSCP)** – Highly respected pentesting certification

 <https://www.offsec.com/courses/pen-200/>

3. **SANS Cybersecurity Training & Certifications**

 <https://www.sans.org/apac/>

4. **CISSP (Certified Information Systems Security Professional)** – For experienced professionals

 <https://www.isc2.org/Certifications/CISSP>

5. **BLT1 (Blue Team Level 1)** – Focused on defensive security & SOC analyst skills

 <https://securityblue.team/certifications/>

Want a Job in Cybersecurity?

-  Most companies require specific certifications for cybersecurity roles.

Top Certifications

1. [CISSP \(Certified Information Systems Security Professional\)](#) – For experienced professionals.
2. [BLT1 \(Blue Team Level 1\)](#) – Defensive security & SOC analyst skills.
3. [OSCP \(Offensive Security Certified Professional\)](#) – Highly respected pentesting certification.

CEH (Certified Ethical Hacker) & Sec+ (CompTIA Security+)

- **Reputation:** Considered **too easy** or even **buyable** by cyber professionals.
- **HR Preference:** Some companies still require them for entry-level roles.

Hands-On Learning Platforms

1. [TryHackMe](#) – Learn cybersecurity concepts, tools & practice challenges.
2. [VulnHub](#) – Free machines for practicing penetration testing.
3. [Hack The Box](#) – Advanced security labs for pentesters.
4. [Cybrary](#) – Free and paid cybersecurity courses.
5. [OverTheWire](#) - Wargames for learning security concepts

Free Cybersecurity Learning – YouTube Channels

1. [NetworkChuck](#) – Fun & engaging cybersecurity tutorials.
2. [Bitten Tech](#) – Hacking & pentesting content.
3. [IppSec](#) – Hack The Box walkthroughs & deep dives.
4. [Hackersploit](#) – Ethical hacking & pentesting.
5. [NullByte](#) – Ethical hacking & security tools.
6. [John Hammond](#) – CTFs, malware analysis & pentesting.
7. [TheCyberMentor](#) – Beginner-friendly hacking tutorials.
8. [TechRaj](#) – Ethical hacking & cybersecurity basics.
9. [HACKADAY](#) – Hardware & software hacking.
10. [LiveOverflow](#) – Deep technical security concepts.
11. [TechChip](#) – Cybersecurity in Hindi.
12. [thenewboston](#) – Programming & security.

13. [Hacking Simplified](#) – Ethical hacking tutorials.
14. [Loi Liang Yang](#) – Cybersecurity & ethical hacking.

Next Steps

- ◆ **Choose a Path** – Offensive security (pentesting), defensive security (SOC analyst), or GRC (Governance, Risk, Compliance).
- ◆ **Start Hands-on Practice** – Use TryHackMe, Hack The Box, and VulnHub to build real skills.
- ◆ **Pick a Certification** – If you're a beginner, start with CompTIA Security+. If you're into pentesting, aim for OSCP.
- ◆ **Build a Portfolio** – Create write-ups of your CTF solutions, bug bounty reports, or personal projects on GitHub.
- ◆ **Stay Updated** – Follow cybersecurity blogs, Twitter accounts, and Reddit communities for the latest trends.

